MOHAMED ATOUI

AI Engineer · Researcher · Builder

London, UK · mohamedelamineatoui@gmail.com · +44 7985 665024

mohamedatoui.uk | LinkedIn | GitHub

SUMMARY

AI engineer in dmg::media's Innovation Team and a First-Class Computer Science & AI graduate from Royal Holloway (2026). Previously co-founded Nordlys Labs, building an LLM platform for **500+ users** with Python/FastAPI services, model routing and production deployment. Experience across applied ML experiments, retrieval evaluation and tool-using agents.

EXPERIENCE

AI Engineer, *dmg::media, Innovation Team* Sep 2026 – Present

Co-organiser, *London Systems Club* Feb 2026 – Present

- Co-organise a London community bringing together systems-level engineers and practitioners to share expertise, exchange ideas, and learn from one another.

Teaching Assistant, *Royal Holloway, University of London* Oct 2025 – May 2026

- Mentored **50+ students** in Object-Oriented Programming and Software Engineering; supported lab sessions with hands-on Java/Python help, and graded assignments with detailed feedback on code quality and engineering standards.

Ambassador, *Thinking About Thinking* Oct 2025 – Present

- Represent the community in London, supporting its events and outreach.

Co-Founder & AI Lead, *Nordlys Labs* Feb 2025 – May 2026

- Built **Hypernova**, a Mixture of Models framework that lets developers compose their own LLM routers across heterogeneous backends. Scored **75.6% on SWE-bench Verified** (Dec 2025), above the strongest published single-model baseline at the time (Claude Opus 4.5, 74.4%), and served 500+ users in production.
- Designed an **intelligent routing layer** that scores prompt complexity and dispatches inference across 30+ models, reducing inference cost 50–60% versus running frontier models directly.
- Built the **production backend** (Python, FastAPI, PostgreSQL, Redis, Docker), the client-facing API, and a **real-time monitoring dashboard** tracking accuracy, latency, and per-model cost. Defined corrective-action workflows when production performance regressed.
- Worked directly with non-technical clients to gather requirements and translate technical work into commercial-grade product surfaces, real-time dashboards, and integration guides.

Product Data Intern, *Stats Perform* Jun 2025 – Oct 2025

- Analysed large-scale sports datasets (NFL, NBA, MLB) using SQL and Python; built feature-engineering pipelines and feature-selection algorithms for production analytics products.
- Modelled revenue projections for AWS marketplace distribution using **Monte-Carlo simulations**, informing go-to-market strategy.
- Worked directly with **enterprise clients** to scope analytics deliverables and presented findings to cross-functional product teams.

Machine Learning Researcher, *Information Security Group (ISG), Royal Holloway* Dec 2024 – Sep 2025

- Researched **adversarial robustness and representation learning** in ML-based perception and intrusion-detection systems for autonomous vehicles, analysing failure modes under attack and distribution shift.
- Designed and trained **CNN and LSTM models** in PyTorch; built an end-to-end evaluation framework with controlled ablations on real-world automotive sensor datasets.
- Co-authored **academic reports and technical whitepapers** documenting methodology, experimental setup, and findings for technical reviewers.

SELECTED PROJECTS

StratigraphicKV, Hybrid KV-Cache Method (Undergraduate FYP) · *Python · PyTorch · HuggingFace Transformers*

- Benchmarked four attention mechanisms (MHA, MQA, GQA, MLA) in iso-parameter 16M-param transformers under controlled experimental conditions; MLA reached **13–26% lower perplexity** than MHA at 72% smaller cache.
- Built **StratigraphicKV**, a layered method combining approaches across the KV-cache literature: classifies attention heads as retrieval or streaming and swaps the streaming cache for O(1) recurrent state. On Llama-3.1-8B: **62% memory reduction** at +0.09 perplexity. Self-directed, topic and direction my own.

ZeroToDiffusion, Diffusion Models From First Principles · *Python · PyTorch*

- From-scratch implementation of **DDPM and a VAE** in PyTorch. Derived the variational lower bound, implemented forward and reverse processes, tested each component (noise schedule, loss, sampler) in isolation before composition.
- Built without library wrappers to develop a working understanding of objective decomposition, inductive biases, and reverse-process parameterisation.

CrossGen, Multi-Stage LLM Reasoning Pipeline · *Python · FastAPI · Claude API · Pydantic*

- Designed a **6-stage agentic reasoning pipeline** (Gentner's Structure-Mapping Theory) that decomposes problems into domain-neutral relations and discovers solutions by mapping structural parallels across 42 curated domains.
- Hybrid deterministic + LLM architecture with mandatory weakness documentation and falsifiable predictions for every output, built for evaluability and reliable production behaviour.

TurboQuant, Vector Quantisation Library (co-built with a friend) · *Zig · SIMD · Python*

- Co-built a Zig implementation of Google's TurboQuant online vector-quantisation algorithm: **~3 bits/dim** ($\approx 6\times$ compression) of float embeddings with near-perfect recall@10. Targets memory-efficient inference under tight resource budgets.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, C, SQL

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, NumPy, Pandas, ONNX

Generative & Representation Learning: Diffusion (DDPM), VAEs, autoregressive transformers, attention mechanisms (MHA/MQA/GQA/MLA), self-supervised methods

Agentic AI & LLMs: LangGraph, LangChain, Claude API, RAG, Sentence Transformers, ChromaDB, multi-agent orchestration, prompt engineering

Research Methods: Ablation studies, systematic benchmarking, adversarial robustness, hypothesis-driven experimentation, technical writing

Infrastructure: Linux, Docker, Git, GitLab CI/CD, FastAPI, AWS, Azure, GPU training

Languages: English (fluent) · French (fluent) · Arabic (native)

EDUCATION

Royal Holloway, University of London Sep 2023 – Jun 2026

BSc (Hons) Computer Science & Artificial Intelligence · First-Class Honours

Quantum Computing · Computational Finance, Machine Learning, Deep Learning, Natural Language Processing, Statistics.

HONORS & AWARDS

- **International Youth Robotics Competition** (2020): placed 1st; the team was honoured by the President of Algeria. I was 14.
- **E-commerce business** (2024): co-founded with two friends after my first year of university; borrowed £1,500 each, ran sourcing/marketing/fulfilment, and turned a profit within three months. Business is still operating today.
- **MoM Framework public talk** (Feb 2026): presented Hypernova's Mixture of Models architecture to a technical audience.